

Validation Plan Overview

Performance and Operational Validation of PeerGFS-Based Active-Active NAS DR
Leading Point | May 2026

1. Purpose

This document provides an overview of our structured validation plan designed to evaluate a PeerGFS-based Active-Active NAS Disaster Recovery (DR) architecture in a production-grade validation environment.

Traditional NAS-based DR solutions commonly suffer from WAN performance degradation, extended RPO windows, inefficient large-file retransmissions, and underutilized passive DR sites. This validation is intended to determine whether PeerGFS can effectively address these limitations in real-world enterprise NAS environments.

The results of this validation will be used in customer demonstrations to objectively validate the operational capabilities, replication performance, and Active-Active DR feasibility of the PeerGFS-based architecture.

2. What We Will Validate

The validation covers six core areas:

Validation Area	Key Verification Points
Near Real-Time Replication	Measure actual replication latency between file modification and DR site synchronization; validate near real-time replication performance and Low RPO operational capability.
Delta-Level Replication	Verify that only modified data blocks (delta changes) are replicated instead of retransmitting entire files, maximizing WAN bandwidth efficiency.
Active-Active File Services	Validate stable bi-directional operations and data consistency during concurrent file creation and modification across both Main and DR sites.
WAN Environment Performance	Validate replication stability and WAN transfer efficiency under degraded network conditions including high latency (50ms), packet loss (1%), and bandwidth constraints (100Mbps).
Failure Recovery and Data Consistency	Verify automatic re-synchronization, operational continuity, and post-recovery data consistency during failure and recovery scenarios.

3. Validation Environment

The validation environment is designed to closely emulate a real-world production deployment for validating real-time replication, WAN performance, failure recovery, and data consistency in an Active-Active NAS DR architecture.

Components: Production-grade NAS, Linux servers (8-core CPU or higher / 32GB RAM or more / SSD 50GB or more / 10Gbps NIC), PeerGFS Agents, Peer Management Center, and WAN network simulation environments.

Dataset: Approximately 10TB / 1.32 million files spanning five file size categories (500KB to 1GB), composed of 99% synthetic data and 1% real-world business files (PDF, JPG, TXT, Office documents).

Initial Data Setup: The full 10TB baseline dataset is established via physical seeding rather than network-based full replication, reflecting real-world migration practices. All subsequent validation focuses on near real-time replication of

delta changes after the initial baseline setup.

WAN Simulation: Linux tc (Traffic Control) is used to emulate realistic WAN conditions including latency, packet loss, and bandwidth limitations.

4. High-Level Methodology

Each validation scenario follows a structured workflow:

- 1. Baseline Measurement:** Network throughput, NAS IOPS, and storage latency are benchmarked using industry-standard tools (iperf3, fio) prior to validation execution.
- 2. Workload Execution:** Scripted file creation, partial overwrite-based modifications, and concurrent bi-directional operations are executed across both Main and DR sites.
- 3. Real-Time Monitoring:** Replication metrics including replication latency, transferred data volume, and concurrent transfer status are monitored in real time through the Peer Management Center.
- 4. Integrity Verification:** Post-replication data consistency is verified using SHA256 checksum comparisons between the Main and DR sites across all validation scenarios.

5. Next Steps

Based on this validation plan, we intend to build a production-grade validation environment and demonstrate the following capabilities in customer-facing demonstrations and technical validation sessions:

- Near real-time replication performance and Low RPO operational capability
- WAN bandwidth optimization through Delta-Level Replication
- Stable Active-Active file services with bi-directional synchronization and data consistency
- Replication stability under degraded WAN conditions including latency and packet loss
- Automatic failure recovery, re-synchronization, and post-recovery data integrity validation

Detailed validation scenarios, operational procedures, and workload scripts are provided in the accompanying validation presentation materials.